Lecture 30.07: Modern Boosting Algorithms — XGBoost, LightGBM, CatBoost











📌 Topic Overview

Comprehensive architectural comparison of modern gradient boosting libraries: XGBoost (2nd-order Taylor expansion & L1/L2 regularization), LightGBM (histogram binning & leaf-wise growth), and CatBoost (categorical encoding & symmetric trees).

📋 Key Learning Objectives

- XGBoost (Extreme Gradient Boosting): Regularization, sparse data handling, and 2nd-order gradients.
- LightGBM (Microsoft): Histogram-based feature binning, leaf-wise tree growth, and memory optimization.
- CatBoost (Yandex): Automatic target encoding for categorical features and symmetric oblivious trees.
- Comprehensive trade-off matrix: Training speed, memory footprint, dataset size suitability, and overfitting risks.

🔍 In-Depth Conceptual Breakdown

- **XGBoost (Extreme Gradient Boosting)**: Gradient boosting ka highly optimized aur scalable version. Kaggle competitions mein dominate karta hai. Isme built-in L1 (Lasso) aur L2 (Ridge) regularization hoti hai jo overfitting rokti hai. Sparse data (missing values) ko natively handle karta hai.

- **LightGBM (Microsoft)**: Microsoft dwara develop kiya gaya lightweight model. Yeh **Histogram-based** technique use karta hai: continuous numerical features ko discrete bins

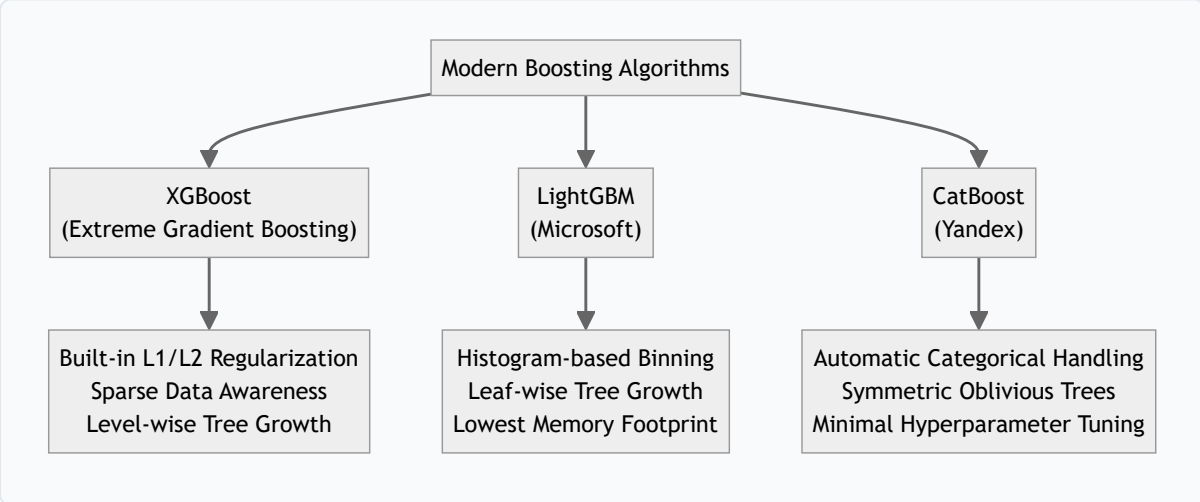
mein convert karke histograms banata hai jisse computation 10x-20x fast ho jati hai. Leaf-wise (best-first) tree growth karta hai. Chote datasets par overfit ho sakta hai, bade datasets ke liye best hai.

- **CatBoost (Yandex)**: Yandex dwara develop kiya gaya algorithm jo **Categorical features** ke liye specially design kiya gaya hai. Isme manual One-Hot Encoding ki zaroorat nahi padti, yeh categorical columns ko automatically process karta hai. Yeh **Symmetric Oblivious Trees** banata hai jisse split balanced rehta hai aur prediction inference speed bohot fast hoti hai.

- **Comparison Table**:

- - **XGBoost**: Best for tabular accuracy, general purpose, robust regularization.
- - **LightGBM**: Fastest training on large datasets ($> 100k$ rows), lowest memory footprint.
- - **CatBoost**: Best out-of-the-box performance when data has many categorical columns.

📊 Architectural Flowchart



🔢 Mathematical Formulation

XGBoost regularized objective function at stage t:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{N} \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^{T} w_j^2 + \alpha \sum_{j=1}^{T} |w_j|$$

where first and second order gradients are:

$$g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)}), \quad h_i = \partial^2_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$$

💻 Implementation

The fully tested and verified standalone implementation is provided in the synchronized Jupyter Notebook.